PPP-S-30D
September 27, 1974
SUPERSEDING
Fed. Spec. PPP-S-30C
April 9, 1973

FEDERAL SPECIFICATION

SACKS, SHIPPING, PAPER (CUSHIONED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cushioned shipping sacks where water resistance and light cushioning effect is required for exterior or interior packaging of items weighing not more than 10 pounds.

1.2 Classification.

1.2.1 Types. The sacks covered by this specification shall be of the following types, as specified (see 6.2).

Type I - Cushioned, Macerated Paper
Type II - Cushioned, Closed Cell Polyethylene
Type III - Cushioned, Polystyrene Foam Beads
Type IV - Cushioned, Laminated Foam/Paper

1.2.2 Dimensions. Cushioned sacks may range from 4 to 20 inches in width and 8 to 40 inches in length, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-P-268 - Paper, Kraft, Wrapping
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-795 - Cushioning Material, Flexible, Cellular, Plastic Film for Packaging Applications
PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical Association of the Pulp and Paper Industry (TAPPI) Publications:

T 402 - Conditioning Paper and Paperboard for Testing
T 404 - Tensile Breaking Strength of Paper and Paperboard
T 410 - Basis Weight of Paper and Paperboard
T 433 - Water Resistance of Paper and Paperboard
T 475 - Bleeding Resistance of Asphalted Paper at Elevated Temperature
T 803 - Puncture and Stiffness Test of Container Board

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Liners. The kraft paper used in the fabrication of the sacks, shall conform to UU-P-268, grade B as follows:

Type I - 60 pound outer liner and 40 pound inner liner (24 x 36-500 basis weight).
Type II - 70 pound outer liner minimum (24 x 36-500 basis weight).
Type III - 60 pound outer liner and 40 pound inner liner (24 x 36-500 basis weight).
Type IV - 60 pound outer liner minimum (24 x 36-500 basis weight).

3.1.2 Cushioning material.

3.1.2.1 Type I. The cushioning material shall be evenly distributed between the exterior and interior sheets of the sack at a thickness of between 1/8 and 3/16 inches and shall have a compression set of not more than 20 percent. The compression properties of the cushioning material shall be determined according to 4.4.3 of this specification.

3.1.2.2 Type II. The cushioning material shall be closed cell plastic film conforming to PPP-C-795, class 1 and shall be 3/16 inch nominal thickness. Compression set shall not be more than 15 percent.

3.1.2.3 Type III. The cushioning material shall be expanded polystyrene foam beads, closely and evenly distributed between the exterior and interior sheets of the sack, a thickness between 1/8 and 3/16 inches and shall have a compression set of not more than 20 percent. The compression properties shall be determined in accordance with 4.4.3.

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3.1.2.4 Type IV. The cushioning material shall be a polyethylene foam, 1/8 inch nominal thickness, have a compression set not more than 20 percent when determined in accordance with 4.4.3.

3.1.3 Liner coating and laminants. The liners shall be treated with coatings or laminants to meet the following requirements:

3.1.3.1 Water resistance. The coated or laminated liners shall have a water resistance sufficient to prevent a definite color change when tested for not less than 1 hour in accordance with 4.4.2.

3.1.3.2 Cushioning binder coating. For type I and III sacks the coating on the kraft liners shall serve to prevent shifting or lumping of the cushioning material when tested in accordance with 4.4.6.

3.1.3.3 Adhesives. For type I and III sacks the coating or laminat shall adhere and seal the exterior and interior sheets at all edges. The seams shall not burst when subjected to the seam strength test specified in 4.4.4.

3.1.3.4 Bleeding. Cushioned shipping sacks shall show no evidence of bleeding or staining when tested as described in 4.4.5.

3.2 Construction.

3.2.1 General requirements, type I and III. The sack shall be of the conventional flat type, with or without a back seam flap of 1-inch minimum. The bottom flap shall not be cushioned and folded to form a seam of 1 to 1-1/2 inches. All seams shall be sealed with a water resistant adhesive conforming to requirements of 3.1.3.3. The sealed edge at the opening of the sack shall not be cushioned, a minimum of 1-inch and a maximum of 2-inches. The sack may be scored 1-inch from the end of the sealed edge to facilitate folding and closing. The material may be serrated at the opening.

3.2.2 General requirements of type II and IV. The sack shall be of the conventional flat type with a bottom fold construction and side seams. The type II sack seams shall first be heat sealed, then folded and glued with a water resistant adhesive. The edge at the opening shall not be cushioned a minimum of 1-inch and a maximum of 2-inches. The sack shall be scored one inch from the opening to facilitate folding for taping or stapling.

3.2.2.1 Additional requirements, type I and III. The walls, of the cushioned sack, shall consist of 2 plies of kraft paper, each coated with a water resistant substance and a cushioning material of specified thickness (see 3.1.2). Type I sacks shall be provided with opening instructions printed on or adjacent to the closing flap so that it is clearly legible when the flap is closed. The printing of waterproof ink shall be in letters not less than 1/4-inch high.

3.2.2.2 Additional requirements, type II and IV. Unless otherwise specified (see 6.2), the sack shall be constructed so that its opening can be closed by heat sealing on conventional heat sealing equipment. Seals, so made, shall be in accordance with 4.4.8. When specified (see 6.2), the type IV sack shall have a top flap closure, effected by a pressure activated adhesive. The adhesive shall be protected with a carrier sheet, capable of being removed merely by stripping without exhibiting adhesive delamination. Adhesive shall be applied across the full width of the flap or overlap area.

3.3 Dimensions. Dimensions of the sack shall be outside measurements, exclusive of flaps when provided (see 3.2.1 and 3.2.2.3), expressed as width by length. Dimensions shall be accurate to within 1/8 inch in width and 1/4 inch in length. See table IV and 6.2.

3.4 Workmanship. Workmanship shall be first class throughout. Sacks shall be uniformly constructed. Free from holes, tears, cuts, splits or other defects which might impair their usefulness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the supplier may utilize his facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified, a lot shall consist of all sacks of one dimension and type submitted for delivery at one time.

4.3 Sampling.

4.3.1 For visual and dimensional inspection. From each lot, a random sample of sacks shall be selected in accordance with MIL-STD-105, for visual and dimensional inspection.

4.3.2 For tests. Samples for test shall be selected from each lot in accordance with table I.

TABLE I

Test Section Number	Sample	Sample Size	Cumulative Sample Size	Cumulative Acceptance Number	Comulative Rejection Number
4.4.1	First	5	5	0	2
	Second	10	15	1	2
4.4.2	First	5	5	0	2
	Second	10	15	1	2
4.4.3	First	5	5	0	2
	Second	10	15	1	2
4.4.4	First	5	5	0	2
	Second	10	15	1	2
4.4.5	First	5	5	0	2
	Second	10	15	1	2
4.4.6	First	5	5	0	2
	Second	10	15	1	2

4.4 Examination. Samples of sacks selected in accordance with 4.3.1 shall be examined for conformity with the requirements, packing and marking provisions of this specification.

4.4.1 Classification of defects. Inspection of sacks shall be conducted as follows:

Major defects, AQL 1.0 percent defective.
1. Material not as specified (see 3.1).
2. Construction not as specified (see 3.2).
3. Dimensions not within tolerance (see 3.3).

Minor defects, AQL 2.5 percent defective.
1. Exterior or interior of sacks torn (see 3.4).
2. Seams not sealed (see 3.2.1).

4.4.2 Water resistance test. Whenever possible, the test sack samples used in the cushioning shifting test shall be used for this test. A six inch square piece of the sack wall (consisting of the 2 kraft sheets, intervening material and coatings) shall be cut from each sample sack. The cut samples shall be carefully taped or stapled, if needed, in a manner that will prevent separation of the sack wall. Each sample shall then be tested for water-resistance in accordance with TAPPI T-433, except that the stipulations concerning the number of the tests to perform and the reporting of average results shall not apply (see 3.1.3.1).

4.4.3 Compression set test. Each sample (table I) consisting of 5 stacked sacks, shall be subjected to a force of 0.2 pounds per square inch, evenly distributed over the surface of the sacks. The thickness of the sacks under this pressure shall be considered the initial uncompressed thickness of the 5 sacks. The sacks shall then be compressed to one-half the initial compressed thickness and maintained at this thickness

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a period of 5 minutes. The sacks shall then be released from all pressure for a period of 2 minutes and finally resubjected to the original force of 0.2 pounds per square inch. The difference between the original thickness of the 5 sacks under a force of 0.2 pounds per square inch (before compression) and the final thickness under a force of 0.2 pounds shall be divided by the original thickness and multiplied by 100, to determine compression set. The test sacks should be allowed to condition without any compression, for a period of 24 hours at 73.40° ± 3.6°F. and 50 ± 2 percent relative humidity before testing (see 3.1.2).

4.4.4 Seam strength test. The sacks (see table I) shall be filled to 3/4 of the total length above the bottom of the sack with dry sand or any other free-flowing material with a density greater than 1-gram per cubic centimeter. The weight of sand or free flowing material added shall not exceed ten pounds. The sacks shall then be subjected to five drops from a height of 36 inch or until failure occurs. The drops shall consist of two drops on the flat sides, one drop on the long side, one drop on the short edge and one drop on the corner of the sack. In all cases, the strength of the side seam and bottom seam or fold shall exceed the burst strength of the bag (see 3.1.3.3).

4.4.5 Bleeding and staining test, type I, III and IV. The sample sacks, as specified in table II, shall be tested for bleeding or staining in accordance with TAPPI-T-475, Bleeding Resistance of Asphalted Paper (see 3.1.3.4).

4.4.6 Cushioning shifting test, type I and III. The test sack sample, as specified in table II, shall be attached rigidly in a verticle position to a conventional vibration machine and vibrated at an amplitude of 1/2 inch and a frequency of 300 c.p.m. for a period of 1 hour. At the end of this test, there shall be no noticeable shifting or lumping of the cushioning material within the sack (see 3.1.3.2).

4.4.7 Heat sealability, type II. The heat sealability test shall be run in accordance with the heat sealed test of MIL-P-116 using a 36 ounce static load.

4.4.8 Examination of preparation for delivery. An examination shall be made to determine that packing and marking as required by section 5 are complied with. Defects shall be as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Marking	Missing; incorrect; illegible; of improper size; location sequence, or method of application.
Materials	Components missing, damaged, or otherwise defective.
Contents	Number of components per container is more or less than required.

4.4.8.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing and strapping shall be in accordance with PPP-B-636.

4.4.8.2 Examination of palletization. An examination shall be made to determine whether palletization as required by section 5 is complied with. Defects shall be as specified in table III. The sample unit shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5, expressed in terms of defects per hundred units.

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TABLE III. Examination of palletization

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps specified.
Weight	Exceeds maximum load.
Marking	Omitted; incorrect; illegible; or improper size; location sequence or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. The sacks shall be packed in a box conforming to PPP-B-636, class weather-resistant, grade V2s. Unless otherwise specified, the quantity per container shall be as specified in table V. The box shall be closed and strapped in accordance with the appendix to the box specification. In lieu of steel strapping, boxes may be reinforced with tape conforming to PPP-T-97, type IV.

5.1.2 Level B. The sacks shall be packed in a box conforming to PPP-B-636, class domestic. Unless otherwise specified, the quantity per shipping container shall be as specified in table V. The box shall be closed in accordance with the appendix to the box specification.

5.1.2.1 When specified (see 6.2) the shipping container shall be V3c, V3s, or V4s fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.1.3 Level C. The sacks shall be packed to insure carrier acceptance and safe delivery at destination at lowest rates, in containers complying with the Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable.

TABLE IV. Quantities for shipment.

Dimensions	Type I No. Bags Per Shipper	Type II No. Bags Per Shipper	Type III No. Bags Per Shipper	Type IV No. Bags Per Shipper
4 x 8	500	500	500	500
5 x 10	250	500	500	500
6 x 10	250	500	500	500
7 1/4 x 12	100	200	500	250
8 1/2 x 12	100	200	250	250
8 1/2 x 14 1/2	100	200	250	250
9 1/2 x 14 1/2	100	200	250	250
10 1/2 x 16	100	100	250	250
12 1/2 x 9	50	100	100	125
14 1/2 x 20	50	100	100	125

5.2 Palletization (military requirements). When specified (see 6.2) sacks, packed as specified, shall be palletized in accordance with load type 1 of MIL-STD-147. Each prepared load shall be banded with primary and secondary straps in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.3 Marking. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable.

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6. NOTES

6.1 Intended use.

6.1.1 Cushioned sacks are frequently used as exterior packaging in the shipment of publications and small parts where light cushioning or water resistance or a combination thereof is required. In some cases, it may be desirable to add reinforcing materials inside the sack to provide protection against puncture or breakage.

6.1.2 Cushioning shipping sacks may also be used for interior packaging of fragile items such as bottled liquids, radio tubes, and testing and laboratory equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, dimension and quantity required.
(c) With or without pressure activated adhesive closure (see 3.2.2.2).
(d) Level of packing required (see 5.1).
(e) When palletization is not required (see 5.2).

MILITARY INTEREST:

Army - E M Q
Navy - S W Sh Y
Air Force - 60

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activity

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